

Smart Sensing Everywhere



RS6241

High Performance Single-Chip 60 GHz MIMO Radar Sensor
with Wireless Connectivity (BLE5.3/802.15.4/Thread/Matter)

Datasheet

Document Sheets: 39

Version: 1.1

Date: 2025-04-14

Revision History

Revision	Notes	Date
1.1	Update Section8.2:Reference Schematic	2025-04-14
1.0	Initial Release	2025-03-24

Contents

Revision History.....	ii
Contents.....	iii
Picture of Contents.....	v
Table of Contents.....	vi
1 Features.....	1
2 Applications.....	1
3 General Description.....	2
3.1 Overview.....	2
3.2 Block Diagram.....	2
4 Pin Configurations.....	3
4.1 Pin Diagrams.....	3
4.2 Pin Descriptions.....	3
4.3 GPIO Multiplexing.....	5
5 Electrical Specifications.....	7
5.1 Absolute Maximum Ratings.....	7
5.2 Recommended Operating Conditions.....	8
5.3 Digital IO Characteristics.....	8
5.4 mmWave Radar RF Specifications.....	8
5.5 Wireless Connectivity RF Specifications.....	9
5.6 Power Specifications.....	10
5.6.1 Power Topologies.....	10
5.6.2 Power Save Modes.....	11
5.6.3 Typical Power Consumption.....	11
6 Function Description.....	12
6.1 Overview.....	12
6.2 mmWave Radar Subsystem.....	12
6.2.1 Receiver.....	12
6.2.2 Transmitter.....	13
6.2.3 Synthesizer.....	13
6.2.4 Chirp Generator.....	13
6.2.5 Baseband.....	14
6.3 2.4GHz Wireless Connectivity Subsystem.....	15
6.4 Application Subsystem.....	15
6.4.1 CPU Processor.....	16
6.4.2 Security Engine.....	17
6.5 Power and Clock Subsystem.....	17
6.5.1 Power Management.....	17
6.5.2 Clock.....	18
6.6 Peripherals.....	19
6.6.1 GPIO.....	19
6.6.2 UART.....	20
6.6.3 QSPI.....	20
6.6.4 I ² C.....	21
6.6.5 PWM.....	21

6.6.6 Audio PWM.....	22
6.6.7 GPADC.....	22
6.6.8 CAN-FD.....	23
6.6.9 LIN.....	23
7 Package Description.....	24
7.1 Package Outline.....	24
7.2 PCB Land Pattern.....	25
7.3 Package Thermal Property.....	26
8 Application Circuit.....	27
8.1 Schematic Notes.....	27
8.2 Reference Schematic.....	28
9 Silkscreen.....	29
10 Shipping package.....	30
11 Reflow Soldering.....	31

Picture of Contents

Figure 3-1	Block Diagram.....	2
Figure 4-1	Pin Diagrams.....	3
Figure 5-1	BOM Optimized (Left) and Power Optimized (Right) Mode.....	11
Figure 6-1	System Block Diagram.....	12
Figure 6-2	Chirp Profile of Frame.....	14
Figure 6-3	2.4GHz Wireless Connectivity Subsystem Diagram.....	15
Figure 6-4	Application Subsystem Diagram.....	16
Figure 6-5	Clock System.....	19
Figure 7-1	Package Outline Diagram.....	24
Figure 7-2	PCB Land Pattern Diagram.....	25
Figure 8-1	Reference Schematic.....	28
Figure 11-1	Recommended reflow soldering heat curve.....	31

Table of Contents

Table 4-1	Pin Descriptions.....	3
Table 4-2	GPIO Multiplexing.....	5
Table 5-1	Absolute Maximum Ratings.....	7
Table 5-2	Recommended Operating Conditions.....	8
Table 5-3	DC Characteristics.....	8
Table 5-4	mmWave Radar RF Specifications.....	8
Table 5-5	Wireless Connectivity RF Specifications.....	9
Table 5-6	Power States.....	11
Table 6-1	Power Management with Block States.....	18
Table 7-1	Thermal Property.....	26
Table 11-1	Recommended thermal distribution parameters.....	31

1 Features

- **FMCW mmWave Radar**

- 57-66 GHz Sweep Frequency Coverage
- 2 Transmit Channels and 4 Receive Channels
- Integrated RFPLL, DPLL, Transmitter, Receiver, ADC and Baseband
- 12.0 dBm Typical Output Power per TX
- 10.5dB Typical NF per RX
- -95 dBc/Hz typical Phase Noise at 1MHz
- 10/20MHz IF Bandwidth
- Up to 800MHz/us Ramp Rate
- HWA for 1D/2D FFT, Static Clutter Remove and CFAR Operations

- **2.4GHz Wireless Connectivity**

- Protocols: Supports Bluetooth LE 5.3, Bluetooth Mesh, IEEE 802.15.4, Thread 1.3, Matter 1.1, and 2.4GHz proprietary
- Bluetooth LE PHY: 1Mbps, 2Mbps, Long Range S2 (500Kbps), S8 (125Kbps)
- TX Max Output Power: 10dBm@BLE mode
- RX Sensitivity: -95dBm @BLE 1Mbps

- **Application System**

- Two 32bit RISC-V CPU Cores with DSP, FPU (up to 256MHz) and I/D cache for Radar Post-processing, Wireless and Application
- Security Engine: Secure Boot, Efuse Key Protection,

Code encryption protection, SRAM and XIP Online Decryption, AES128/SHA/CRC/MD5/RSA/ECC/TRNG

- **Memory**

- On Chip 551KB SRAM
- External PSRAM support (up to 16MB)

- **Host and Device Interface**

- x3 UART
- x2 QSPI with XIP
- x2 I²C
- x8 PWM
- X10 GPADC
- x1 CAN-FD
- x1 LIN

- **Clock Source**

- 32/40MHz Crystal Oscillator for Clock Reference
- Optional 32kHz Crystal for Low Power Operation

- **Power Management**

- Single Power Supply from 1.7V to 5.5V
- Built-in LDO Network for Enhanced PSRR
- BOM-Optimized and Power-Optimized Modes

- **Low Power Property**

- Shutdown, Standby, Idle and Active (Sensing, Processing and Connecting) Mode
- Fast Wakeup from Standby to Active

- **Package**

- FCCSP: 6.7 x 6.7 mm², with 94 pins

- **Temperature Range**

- Operating Junction Temperature: -40°C to 125°C

2 Applications

- Motion /Micro Motion/Presence detection
- Vital signs/health monitoring
- Occupancy detection
- People tracking/counting
- Posture and fall detection
- Industrial detection
- Drone/robot obstacle avoidance
- In-cabin sensing for left behind (LBC) systems

3 General Description

3.1 Overview

The RS6241 is a single chip 60GHz 2T4R FMCW MIMO radar sensor with 2.4GHz wireless connectivity (BLE5.3/802.15.4/Thread/Matter) inside. The device is portioned into four subsystems, shown in Figure below:

- **mmWave Radar Subsystem:** This block includes all high-performance radar RF/analog and baseband. It also includes the chirp generator and tx-to-rx finite state machine. HWA is included in baseband, used to offloading specified radar processing such as 1D/2D FFT and CFAR.
- **2.4GHz Wireless Connectivity Subsystem:** This block contains whole 2.4G multi-protocol wireless RF, Mod/Demod, Baseband Protocol, and the data interacts with application CPU by internal data bus.
- **Application Subsystem:** It mainly includes two 32bit RISC-V CPU cores with DSP, FPU and I/D cache. CPU connects SRAM, peripheral interface, and other subsystems by AHB and APB bus. RISC-V CPU is used to run radar post-processing algorithm, wireless protocol and other system tasks.
- **Power and Clock Subsystem:** This chip is complex single chip SoC with multiple function systems, we use single power and clock subsystem to do centralized management. It includes several clock sources (including LSI, MSI, Xtal Oscillator and SPLL), power blocks(LDOs, BG and Clamp) and temperature sensor.

RS6241 is specifically designed for low power, high accuracy radar sense system applications. It contains four main power modes for low power management: shutdown, standby, idle and active. Its configuration and data acquisition are enabled with UART/SPI/I2C/CAN-FD/LIN digital interface with other devices.

3.2 Block Diagram

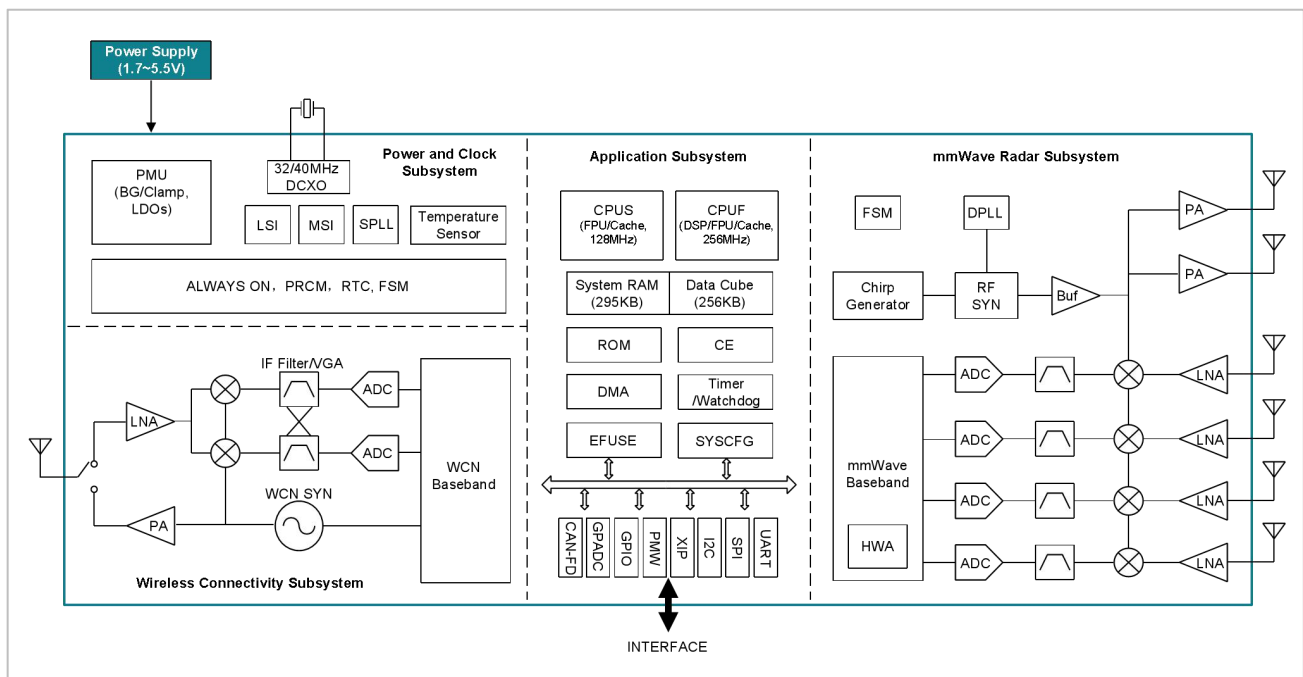


Figure 3-1 Block Diagram

4 Pin Configurations

4.1 Pin Diagrams

TOP VIEW

	1	2	3	4	5	6	7	8	9	10
A	GND	ANT	GND	TX1	GND	TX0	GND	GND	VRF	GND
B	GND	GND	GND		GND		GND	GND	GND	GND
C	XL2	GND	GND	GND	GND	GND	V13T	GND		RX0
D	XL1	GND	GND	V13M	GND	GND	GND	GND	GND	GND
E	PA12	GND	GND	V18C	GND	GND	GND	GND		RX1
F	PA11	PA13	PA14	PA15	V18A	V13R	GND	GND	GND	GND
G	PA2	PA3	PA8	PA9	PA10	PA7	GND	GND		RX2
H	VDDIO	PA1	PA4	PA5	PA6	PA0	NRST	GND	GND	GND
K	VEXT	VCC	VINT	PB0	PB5	PB3	PB2	GND		RX3
L	GND	VCC	VDIG	PB1	PB4	PB6	PB7	GND	GND	GND

Figure 4-1 Pin Diagrams

4.2 Pin Descriptions

Table 4-1 Pin Descriptions

Pin Name	I/O	Pin No.	Pin Description
VCC	Power In	K2,L2	PMU power supply, 1.7~5.5V
VINT	Power Out	K3	PMU INT LDO output, Default 1.8V, 0.9~2.5V/0.1V step
VEXT	Power Out	K1	PMU EXT LDO output, Default 3.3V, 1.6~3.3V/0.1V step
VDIG	Power Out	L3	PMU DIG LDO output, Default 1.15V, 0.65~1.35V/0.05V step
VDDIO	Power In	H1	GPIO Power supply, 1.7~3.6V
V18A	Power In	F5	MMW Analog power supply, 1.5~2.5V
V18C	Power In	E4	MMW Analog power supply, 1.5~2.5V
V13T	Power In	C7	MMW TX power supply, 1.2~2.5V
V13R	Power In	F6	MMW RX power supply, 1.2~2.5V
VRF	Power Out	A9	MMW RF LDO output, default 1.0V
V13M	Power In	D4	2.4G wireless analog power supply, 1.2~2.5V

XL1	Analog	D1	Crystal oscillator output
XL2	Analog	C1	Crystal oscillator input
ANT	RF	A2	2.4G RF input and output
TX0	RF	A6	RADAR RF output0
TX1	RF	A4	RADAR RF output1
RX0	RF	C10	RADAR RF input0
RX1	RF	E10	RADAR RF input1
RX2	RF	G10	RADAR RF input2
RX3	RF	K10	RADAR RF input3
GND	Ground	/	Chip ground
NRST	In	H7	Chip reset, need feed VDDIO to power supply
PA0	In/Out	H6	Programmable GPIO input/output, Wakeup IO
PA1	In/Out	H2	Programmable GPIO input/output, Wakeup IO
PA2	In/Out	G1	Programmable GPIO input/output, Wakeup IO
PA3	In/Out	G2	Programmable GPIO input/output, Wakeup IO
PA4	In/Out	H3	Programmable GPIO input/output, Wakeup IO
PA5	In/Out	H4	Programmable GPIO input/output, Wakeup IO
PA6	In/Out	H5	Programmable GPIO input/output, Wakeup IO
PA7	In/Out	G6	Programmable GPIO input/output, Wakeup IO
PA8	In/Out	G3	Programmable GPIO input/output, Wakeup IO
PA9	In/Out	G4	Programmable GPIO input/output, Wakeup IO
PA10	In/Out	G5	Programmable GPIO input/output, Wakeup IO
PA11	In/Out	F1	Programmable GPIO input/output, Wakeup IO
PA12	In/Out	E1	Programmable GPIO input/output, Wakeup IO
PA13	In/Out	F2	Programmable GPIO input/output, Wakeup IO
PA14	In/Out	F3	Programmable GPIO input/output, Wakeup IO
PA15	In/Out	F4	Programmable GPIO input/output, Wakeup IO
PB0	In/Out	K4	Programmable GPIO input/output, Wakeup IO
PB1	In/Out	L4	Programmable GPIO input/output, Wakeup IO
PB2	In/Out	K7	Programmable GPIO input/output
PB3	In/Out	K6	Programmable GPIO input/output
PB4	In/Out	L5	Programmable GPIO input/output
PB5	In/Out	K5	Programmable GPIO input/output
PB6	In/Out	L6	Programmable GPIO input/output

PB7	In/Out	L7	Programmable GPIO input/output
-----	--------	----	--------------------------------

4.3 GPIO Multiplexing

Table 4-2 GPIO Multiplexing

GPIO Name	Func2	Func3	Func4	Func5	Func 6	Func7	Func after rst
PA0/WIO0	SPI0_CS	UART1_TX	CANFD_TX		PWM0		OFF/No-Pull
PA1/WIO1	SPI0_CLK	UART1_RX	CANFD_RX		PWM1		OFF/No-Pull
PA2/WIO2	SPI0_MOSI	UART1_RTS	I2C0_SDA	UART2_TX/LIN_TX	PWM2		OFF/No-Pull
PA3/WIO3	SPI0_MISO	UART1_CTS	I2C0_SCL	UART2_RX/LIN_RX	PWM3		OFF/No-Pull
PA4/WIO4	SPI0_HOLD	I2C0_SDA			PWM4		OFF/No-Pull
PA5/WIO5	SPI0_WP	I2C0_SCL			PWM5		OFF/No-Pull
PA6/WIO6	UART1_RTS	UART2_TX	I2C1_SDA		PWM6	GPADC6	OFF/No-Pull
PA7/WIO7	UART1_CTS	UART2_RX	I2C1_SCL		PWM7	GPADC7	OFF/No-Pull
PA8/WIO8/LXL1 ¹	UART1_TX	I2C1_SCL			PWM0	GPADC0	OFF/No-Pull
PA9/WIO9/LXL2 ¹	UART1_RX	I2C1_SDA			PWM1	GPADC1	OFF/No-Pull
PA10/WIO10	I2C1_SDA	CANFD_TX			PWM2	GPADC2	OFF/No-Pull
PA11/WIO11	I2C1_SCL	CANFD_RX			PWM3	GPADC3	OFF/No-Pull
PA12/WIO12	UART2_RX	UART1_RTS			PWM4	GPADC4	OFF/No-Pull
PA13/WIO13/TEST ²	UART2_TX	MCO_CLK			PWM5	GPADC5	IN/UP
PA14/WIO14	TWIO_SDA	UART2_RX		CANFD_TX	PWM6		OFF/No-Pull
PA15/WIO15	TWIO_SCL	UART2_TX		CANFD_RX	PWM7		OFF/No-Pull
PB0/WIO16	UART0_TX				PWM0		OFF/UP
PB1/WIO17	UART0_RX				PWM1		OFF/UP
PB2	UART0_RTS	FC_WP	SPI0_WP		PWM2		OFF/UP
PB3	UART0_CTS	FC_HOLD	SPI0_HOLD		PWM3		OFF/UP
PB4	FC_CS	UART2_RX	SPI0_CS		PWM4		OFF/No-Pull
PB5	FC_CLK	UART2_TX	SPI0_CLK		PWM5		OFF/No-Pull
PB6	FC_MOSI	I2C0_SDA	SPI0_MOSI		PWM6		OFF/No-Pull
PB7	FC_MISO	I2C0_SCL	SPI0_MISO		PWM7		OFF/No-Pull

1. LXL1 and LXL2 are 32K external crystal pins, they share the same ball with PA8 and PA9, when used as analog pin, GPIO function should be disabled.
2. When chip is first power on or after reset, PA13 will be configured to input and pull-up mode, if it is tied to GND then will go into ATE test mode.

5 Electrical Specifications

5.1 Absolute Maximum Ratings

Table 5-1 Absolute Maximum Ratings

Parameters ¹		Min.	Max.	Unit
VCC	PMU power supply	-0.30	6.50	V
VINT	PMU INT LDO output	-0.30	3.80	V
VEXT	PMU EXT LDO output	-0.30	3.80	V
VDIG	PMU DIG LDO output	-0.30	1.40	V
VDDIO	GPIO Power supply	-0.30	3.80	V
V18A	MMW Analog power supply	-0.30	3.80	V
V18C	MMW Analog power supply	-0.30	3.80	V
V13T	MMW TX power supply	-0.30	3.80	V
V13R	MMW RX power supply	-0.30	3.80	V
VRF	MMW RF LDO output	-0.30	1.40	V
V13M	2.4G wireless analog power supply	-0.30	3.80	V
V _{XL}	Crystal oscillator input and output voltage	-0.30	2.70	V
P _{ANT}	2.4G RF input power	-	10	dBm
V _{IO}	GPIO voltage for input and output	-0.30	3.80	V
I _{IO}	GPIO In/Out current for input and output	-100	100	mA
T _{opr}	Ambient Operating Temperature	-40	105	°C
T _{junction}	Operating junction temperature range	-40	125	°C
T _{stg}	Storage temperature range after soldered onto PC board	-55	150	°C
V _{HBM}	Human-body model (HBM), ANSI/ESDA/JEDEC JS-001	-2000	2000	V
V _{CDM}	Charged-device model (CDM), ANSI/ESDA/JEDEC JS-002 ²	-500	500	V

1. Stresses beyond those listed under Absolute Maximum Ratings may cause permanent damage to the device. These are stress ratings only, and functional operation of the device at these or any other conditions beyond those indicated under Recommended Operating Conditions is not implied. Exposure to absolute-maximum-rated conditions for extended periods may affect device reliability.
2. Corner pins are rated as ± 750 V.

5.2 Recommended Operating Conditions

Table 5-2 Recommended Operating Conditions

Parameters		Min.	Max.	Unit
VCC	PMU power supply	1.70	5.50	V
VINT	PMU INT LDO output	1.20	2.50	V
VEXT	PMU EXT LDO output	1.70	3.60	V
VDIG	PMU DIG LDO output	0.65	1.25	V
VDDIO	GPIO Power supply	1.70	3.60	V
V18A	MMW Analog power supply	1.70	2.50	V
V18C	MMW Analog power supply	1.70	2.50	V
V13T	MMW TX power supply	1.20	2.50	V
V13R	MMW RX power supply	1.20	2.50	V
V13M	2.4G wireless analog power supply	1.20	2.50	V
VRF	MMW RF LDO output	0.80	1.20	V

5.3 Digital IO Characteristics

Table 5-3 DC Characteristics

Parameters		Min.	Max.	Unit
VIL	Input Low Voltage	-0.3	0.35*VDDIO	V
VIH	Input High Voltage	0.65*VDDIO	1.1*VDDIO	V
VOL	Output Low Voltage	-0.3	0.1*VDDIO	V
VOH	Output High Voltage	0.9*VDDIO	1.1*VDDIO	V
RPU	Input Pull-up Resistance@VDDIO=3.3V	35	75	Kohm
	Input Pull-up Resistance@VDDIO=1.8V	70	150	Kohm
RPD	Input Pull-down Resistance@VDDIO=3.3V	30	75	Kohm
	Input Pull-down Resistance@VDDIO=1.8V	60	150	Kohm

5.4 mmWave Radar RF Specifications

Table 5-4 mmWave Radar RF Specifications

Parameters		Min.	Typ.	Max.	Unit
Receiver	Noise figure	-	10.0	-	dB
	Maximum gain	-	51	-	dB

	Gain range		-	24	-	dB
	Gain step size		-	6	-	dB
	IF bandwidth ¹		10	-	20	MHz
	ADC resolution		-	12	-	Bit
	Gain mismatch variation (over temperature)		-	+/-0.5	-	dB
	Phase mismatch variation (over temperature)		-	+/-3	-	°
Transmitter	Max Output Power		-	12.0	-	dBm
	Power backoff range		-	20	-	dB
	Temperature sensor accuracy		-	+/-3	-	°C
Synthesizer	Frequency range		57	-	66	GHz
	Continuous linear sweep frequency bandwidth		-	-	8	GHz
	Ramp rate		-	-	800	MHz/us
	Phase noise at 1MHz offset ²	2GHz sweep BW	-	-95	-	dBc/Hz
		6GH sweep BW	-	-92	-	dBc/Hz
		8GH sweep BW	-	-89	-	dBc/Hz

1. IF bandwidth can be configured to 10M/20M mode, and LPF is also be configured with IF bandwidth.
2. Data is measured at 60GHz with DCXO at 40MHz at typical 25°C. MMW Synthesizer can be configured to high BW mode (6GHz sweep bandwidth) or high performance mode (better phase noise, 2GHz sweep bandwidth).

5.5 Wireless Connectivity RF Specifications

Table 5-5 Wireless Connectivity RF Specifications

Parameters		Min.	Typ.	Max.	Unit
Receiver	Frequency range	2400	-	2484	MHz
	RX sensitivity@BLE-1Mbps	-	-95	-	dBm
	RX sensitivity@BLE-2Mbps	-	-92	-	dBm
	RX sensitivity@BLE-1Mbps, Coded2	-	-99	-	dBm
	RX sensitivity@BLE-1Mbps, Coded8	-	-101	-	dBm
	RX sensitivity@IEEE802.15.4	-	-100	-	dBm
	Maximum receiving power	-	0	-	dBm
	BLE C/I@Co-channel	-	7	-	dB
	BLE C/I@+1MHz ¹	-	-2	-	dB
	BLE C/I@+2MHz	-	-37	-	dB
	BLE C/I@+3MHz	-	-41	-	dB
	BLE C/I@+4MHz	-	-41	-	dB

	BLE C/I@-1MHz		-	2.5	-	dB
	BLE C/I@-2MHz		-	-19	-	dB
	BLE C/I@-3MHz		-	-25	-	dB
	BLE C/I@-4MHz		-	-37	-	dB
	IEEE802.15.4 C/I@+10MHz ²		-	-59	-	dB
	IEEE802.15.4 C/I@+5MHz		-	-54	-	dB
	IEEE802.15.4 C/I@-5MHz		-	-40	-	dB
	IEEE802.15.4 C/I@-10MHz		-	-60	-	dB
Transmitter	Frequency range		2400	-	2484	MHz
	FSK data rate	BLE	-	1.0	2.0	Mbps
		802.15.4	-	2.0	-	Mbps
	Modulation deviation	BLE 1Mbps	225	250	275	kHz
		BLE 2Mbps	450	500	550	kHz
		IEEE 802.15.4	-	500	-	kHz
	Output power range		-20	-	11.5	dBm
	In band spur. emission at 1Mbps@2M offset ³		-	-50	-	dBm
	In band spur. emission at 1Mbps@>=3M offset ³		-	-54	-	dBm
	In band spurious emission at 2Mbps@4M offset ³		-	-55	-	dBm
	In band spur. emission at 2Mbps@5M offset ³		-	-57	-	dBm
	In band spur. emission at 2Mbps@>=6M offset ³		-	-60	-	dBm
	Out band spur. emission at 4dBm output power	<2.4GHz	-	-56	-	dBm
		2.48~12GHz (wo harmonics)	-	-56	-	dBm
	Harmonics at 4dBm output power	2 nd	-	-41	-	dBm
		3 th	-	-43	-	dBm

1. The desired signal is at -67dBm;
2. The desired signal is at -82dBm;
3. Output power at 2dBm;

5.6 Power Specifications

5.6.1 Power Topologies

The chip supports two unique power topologies for BOM Optimized and Power Optimized modes. BOM Optimized Mode uses internal LDOs of PMU only to reduce extra device cost. Power Optimized Mode uses external DCDC to feed to V13 to reduce power consumption. The main power supply VCC can be provided from 1.7 to 5.5V depends on the system request.

VDDIO can be provide by EXT LDO inside chip or from outside. When Power Optimized Mode is selected, the EXT LDO is provided for V18 by default and VDDIO will be fed by external power supply. When BOM Optimized Mode is selected, the EXT LDO is provided to VDDIO by default, and also can be used for V18 power supply.

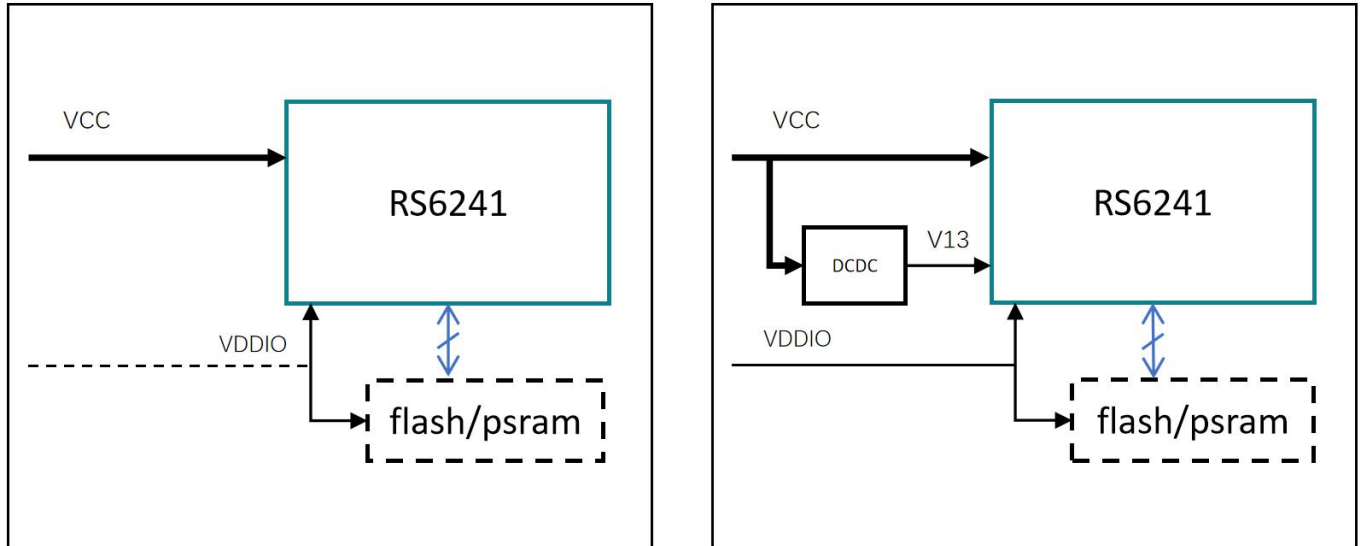


Figure 5-1 BOM Optimized (Left) and Power Optimized (Right) Mode

5.6.2 Power Save Modes

Table 5-6 Power States

Parameters		
Active	Sensing	Radar power state when RF/chirp is ongoing
	Processing	Radar data processing state when RF on is finished
	Connecting	Wireless active power state when RF/PHY /MAC is on
Idle		Power state during inter-frame/inter-chirp idle time
Standby		Lowest power state of the chip(only RTC power domain is activated) where the contents can be retained by retention SRAM and chip can fast wakeup to active
Shutdown		Chip is power down by NRST pin setting to low

5.6.3 Typical Power Consumption

TBD.

6 Function Description

6.1 Overview

The RS6241 is a fully integrated SoC which includes mmWave radar subsystem, programmable CPU application subsystem, 2.4GHz multi-protocol wireless connectivity subsystem and power/clock subsystem, as shown in Figure 6-1 System Block Diagram. It is designed for Artificial Intelligence and Internet of Things (AIOT) sensing applications, such as motion sensing, gesturing, automation sensing and building energy saving.

The SoC is optimized for low-power operation by using several low-power states and fast wake-up times from hardware to software. Multiple power domains and clocks can be shut down individually. The application subsystem, mmWave subsystem and wireless subsystem can be put into low-power states independently, supporting a variety of application use cases.

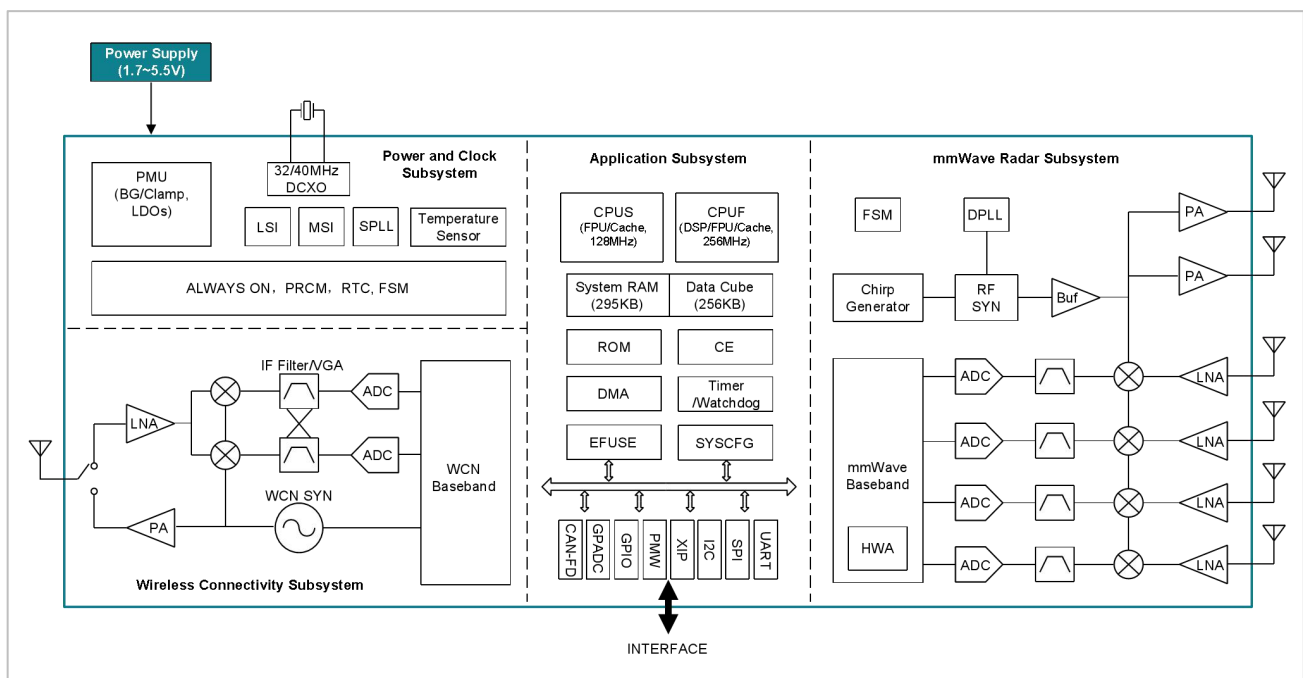


Figure 6-1 System Block Diagram

6.2 mmWave Radar Subsystem

The mmWave radar subsystem includes receiver, transmitter, synthesizer, chirp generator and baseband. It is designed to transmit and receive FMCW radar signal, and further to transfer radar signal to desired target information, such as range, velocity, angle and micro-motion.

6.2.1 Receiver

Receiver block consists of four parallel channels. A single receive channel consists of an LNA, mixer, IF filtering, ADC conversion, and decimation. All four receive channels can either operate simultaneously or can be powered down individually based on system power needs and application design. The RS6241 band-pass IF chain has configurable lower cutoff frequencies above 500 kHz and can support bandwidths up to 20 MHz.

6.2.2 Transmitter

Transmitter block consists of two transmit chain with amplitude control for power back-off. A power stage driving by driver stage to get high power(>10dBm) for long range detection.

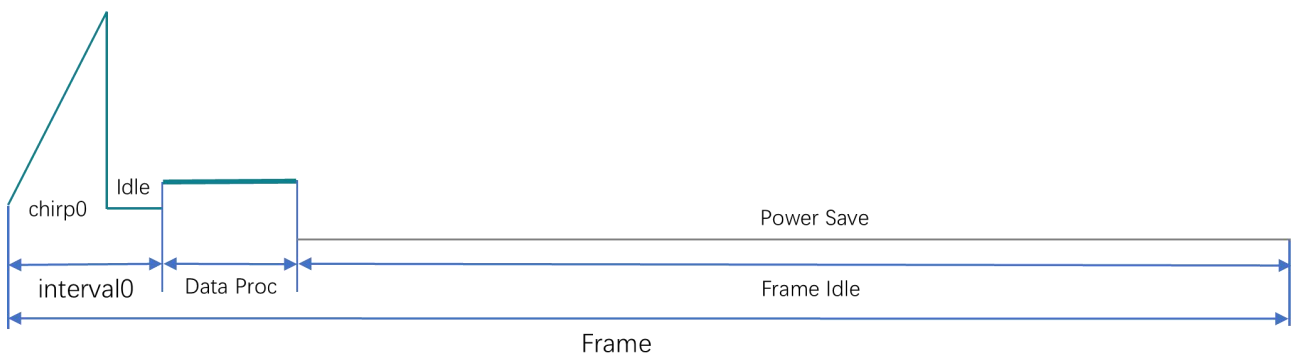
6.2.3 Synthesizer

Synthesizer block generates 57 to 66 GHz from an input reference from a crystal. It has a built-in oscillator circuit followed by a DPLL and a RF synthesizer circuit. The output of the RF synthesizer is then processed by an X4 multiplier to create the required frequency in the 57 to 66 GHz spectrum. The RF synthesizer output is modulated by the timing engine block to create the required waveform for effective sensor operation.

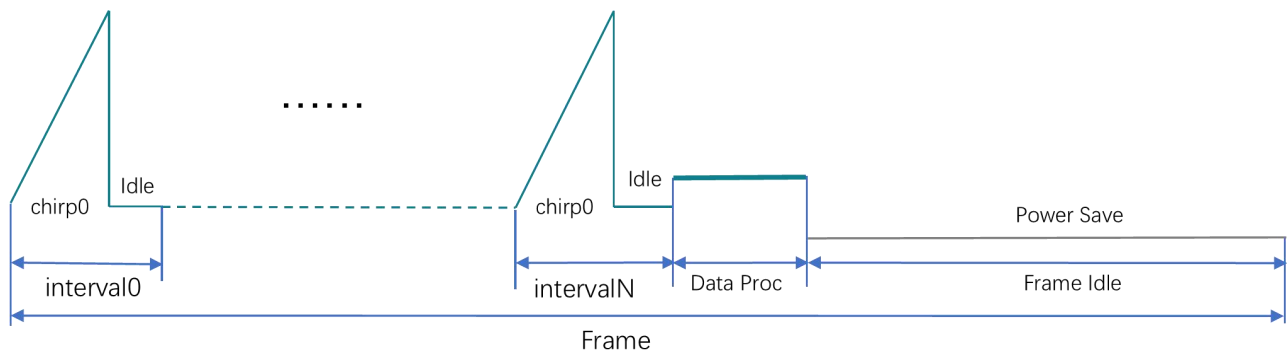
6.2.4 Chirp Generator

Chirp generator block is used to generate different kinds of FMCW chirps by supporting for chirps, intervals and frames. The device only support ramp up FMCW mode to simplified system. The chirp profile is shown in Figure 6-2 , We can set three different frame mode:

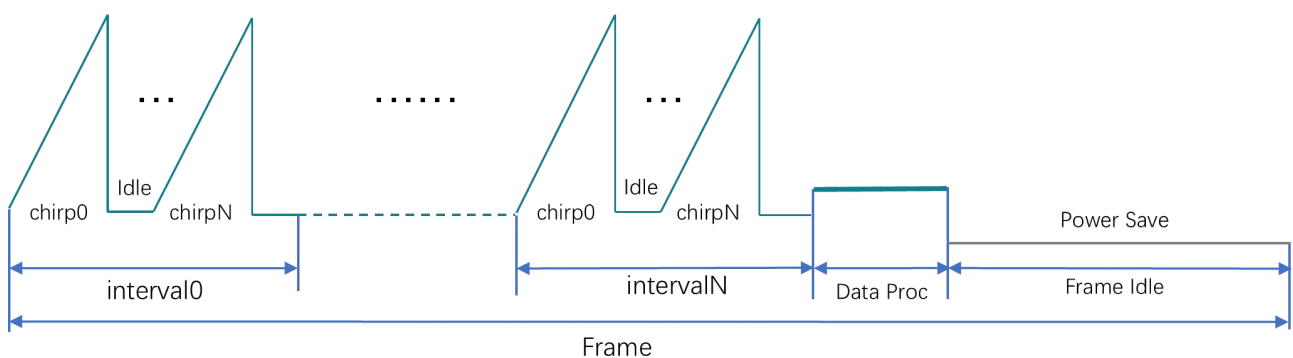
- (a) **Single-Interval and Single-Chirp Frame Mode:** Only one chirp in a frame, 1D-FFT with range information is obtained only, and system can go into Standby at frame idle time to significantly save power;
- (b) **Multi-Interval with Single Chirp in Interval Frame Mode:** One frame with multiple interval for 2D-FFT calculation to get both range and velocity information, interval number represents the Doppler number;
- (c) **Multi-Interval with Multi-Chirp Accumulation Frame Mode:** Based on the above mode (b), the chirp number can be set to $N > 1$ at each interval, which will set chirp to accumulation (averaging) mode every interval. Its possible to across closely spaced chirps to reduce memory requirement.



(a) Single-Interval and Single-Chirp Frame Mode



(b) Multi-Interval with Single Chirp in Interval Frame Mode



(c) Multi-Interval with Multi-Chirp Accumulation Frame Mode

Figure 6-2 Chirp Profile of Frame

6.2.5 Baseband

Baseband block consists of several signal processing units, such as digital filter, frame control, FIFO and hardware accelerator(HWA). HWA is used to offloading common radar processing such as FFT, Constant False Alarm rate (CFAR) and pre-processing. HWA features are as follows:

- Supports 2^N ($N=2,3,... 10$) complex FFT of any length input and up to 1024 times adaptive gain scaling, including Rectangular Windows, Hanning Windows, Hamming Windows, Blackman Windows and etc;
- Supports hardware CFAR up to 512 point clouds;
- Supports built in data pre-processing functions, including DC bias elimination, phase alignment between Rx antennas, decoupling between Rx antennas, phase alignment between Tx antennas, static clutter remove and other functions;
- Supports the chirp accumulation(averaging) mechanism, averaging up to 128 chirps;
- Supports Chirp interference detection, abnormal peak detection on time-domain waveform, and zero setting;
- Data output is configurable with flexible allocation of distance FFT and Doppler FFT points. It supports three types of output waveform: time-domain waveform, distance spectrum, and distance Doppler spectrum.

6.3 2.4GHz Wireless Connectivity Subsystem

The RS6241 includes one 2.4GHz wireless connectivity subsystem, which is designed ready for the major feature advancements of Bluetooth®5 and takes advantage of Bluetooth 5's increased performance capabilities which include long range and high throughput modes. The 802.15.4 PHY and MAC layers are supported native on the RS6241. This allows RS6241 to be used in a wide range of home and industrial sensor network applications as it supports two of the most popular wireless sensor standards in use today, Bluetooth low energy and 802.15.4 derivatives. This adds to the already existing radio support for Bluetooth low energy, Thread/Matter and 2.4GHz for proprietary.

The wireless connectivity subsystem contains one 2.4G RF radio transceiver, BLE5 PHY, 802.15.4 PHY, BLE5 Link Layer and 802.15.4 MAC. And the other relative protocol is driven by CPU system. The system diagram is shown in Figure 6-3 2.4GHz Wireless Connectivity Subsystem Diagram.

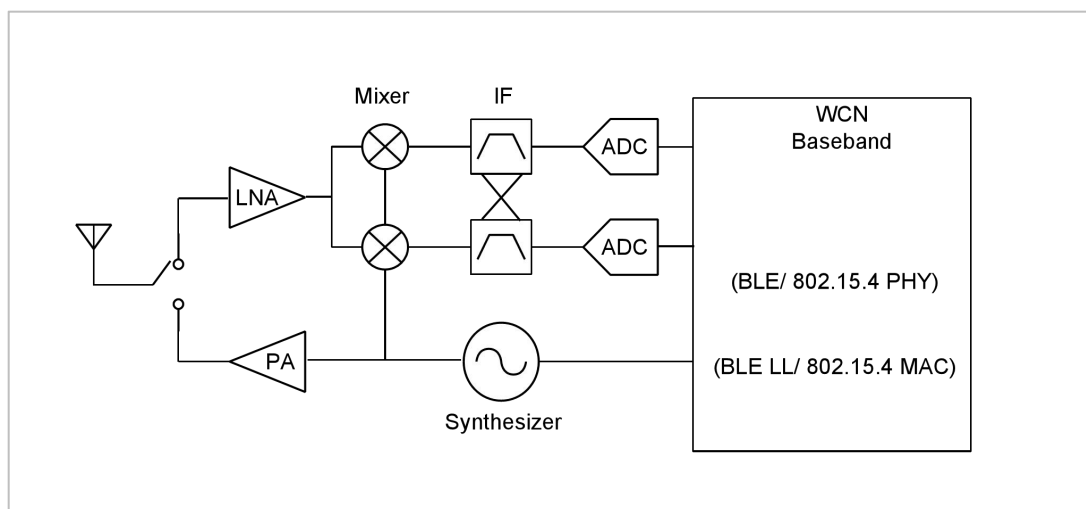


Figure 6-3 2.4GHz Wireless Connectivity Subsystem Diagram

The RF transceiver works in the whole 2.4GHz ISM band, which contains a fully integrate RF synthesizer, a power amplifier, a modulator and a receiver. The RF synthesizer is a Sigma-Delta Fractional-N PLL with LC VCO, while transmit modulation is directly applied to the loop and IQ signal is generate to RX mixer in the receive mode. The receiver includes one Low Noise Amplifier(LNA), IQ mixer and poly-phase IF filter with VGA function, and the Analog-to-Digital Converter(ADC). A switch mode power amplifier is used to optimize PA efficiency, and transmit output power from -20dBm to 10dBm. The RF transceiver also includes several auto calibration machines such as AFC, RX filter tuning, RX offset calibration and PLL bandwidth calibration.

The baseband contains BLE/802.15.4 PHY, BLE Link Layer and 802.15.4 MAC. It supports 125Kbps, 500Kbps, 1Mbps or 2Mbps for BLE5 and 250Kbps for IEEE802.15.4. The baseband also contains hardware logic to perform fast AGC control, access code correction, CRC checking, data whitening, encryption/decryption and frequency hopping logic.

6.4 Application Subsystem

The Application Subsystem is the brain of the chip and controls all the device peripherals and house-keeping activities of the device. It contains 32-bit RISC-V CPU processor, Security Engine, ROM and RAM memory, AHB/APB bus, and associated peripherals and house-keeping components such as DMA and Peripherals (UART, SPI, I²C, GPIO, PWM, and others). The block diagram is shown in Figure 6-4.

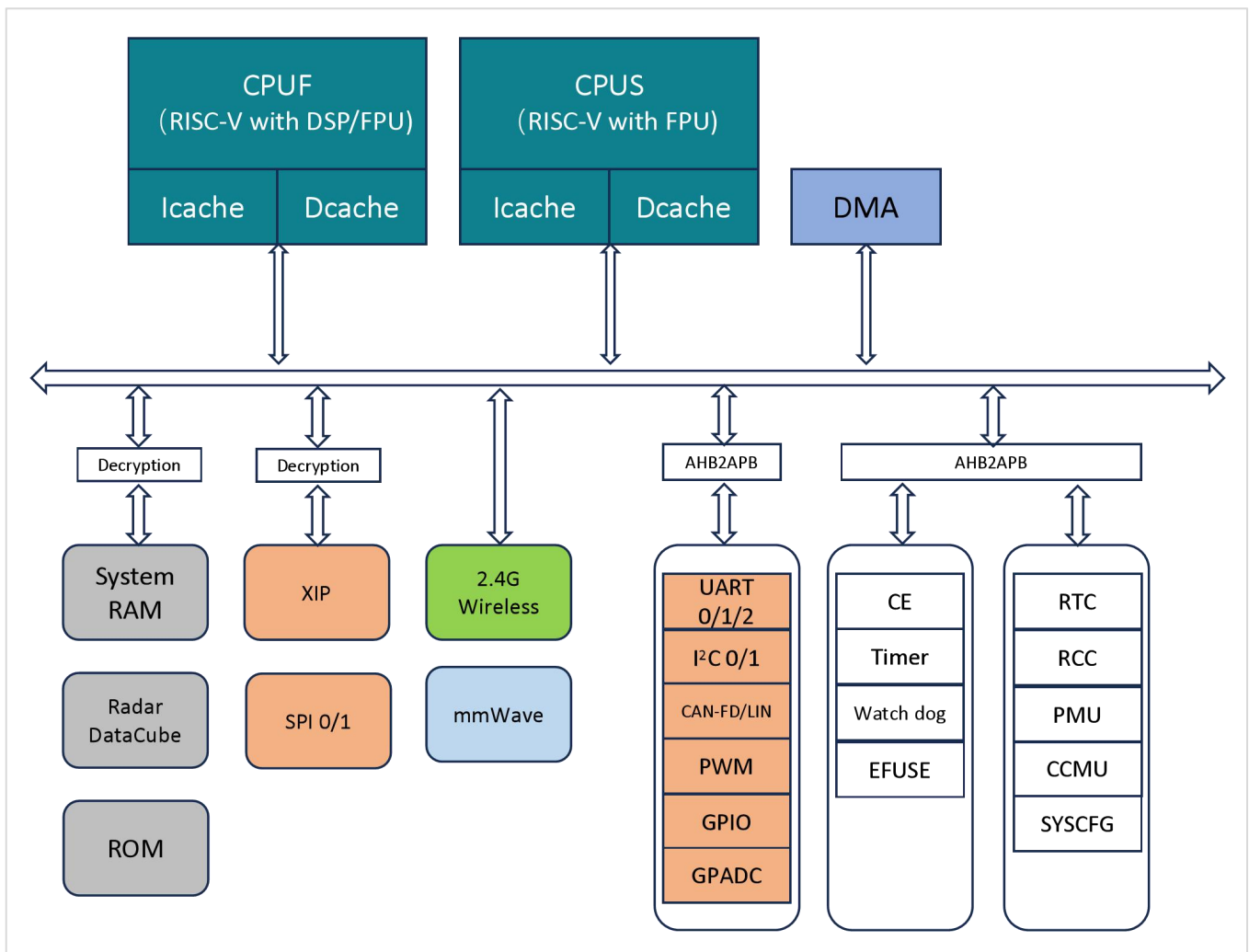


Figure 6-4 Application Subsystem Diagram

6.4.1 CPU Processor

The 32-bit RISC-V CPU processor is clocked with a maximum operating frequency of 128 MHz. User applications executing on this processor control the overall operation of the device, including radar control through well-defined API messages, radar signal processing (assisted by the radar hardware accelerator), 2.4GHz wireless protocol, and peripherals for external interfaces. The processor provides a low-cost platform that meets the needs of minimal memory implementation, reduced pin count, and low power consumption. It has the following features:

- 32-bit RSIC Core, RISC-V 32b/16b variable length ISA
- RV32IMAC compatible
- RV32[D][F]standard extensions for single-precision floating-point, Compatible with IEEE-754 and DSP
- Machine mode + user mode
- Improved 5 stage, out-of-order completion
- I & D caches

- Two-level dynamic branch prediction
- 8 channels Physical Memory Protection (PMP) for memory protection features
- Core Local Interrupt Controller (CLIC) to implement interrupt processing
- Support 2k-bit Branch History Table (BHT)
- Support 16-entry Branch Target Buffer (BTB)
- Three 32-bit AMBA3 AHB-Lite interfaces: instruction, data and system bus
- Debug Interfaces support 2-wires SWD and 5-wires JTAG

6.4.2 Security Engine

The RS6241 provides fully security protection for IOT application from boot, key, code and network. It has the following features:

- Secure Boot: ROM supports secure boot to check the integrity and authenticity of the firmware before executing it
- Efuse Key Protection: the secure key in the Efuse is prevent to illegally read
- SRAM and XIP Decryption on the fly: Bus supports Decryption on the fly from external RAM and Flash
- Crypto Engine:
 - AES supports 128, 192, 256 three key lengths and ECB, CBC, CFB, OFB, CTR, GCM five modes
 - SHA1, SHA224 and SHA256
 - RSA supports up to 4096
 - ECC supports up to 521
- Hardware TRNG

6.5 Power and Clock Subsystem

6.5.1 Power Management

A single 1.7~5.5V power supply is required for the RS6241. It could be from an AC-DC converter, USB to supply to 5V or a DC-DC converter to convert higher voltage supply to 3.3V or even lower. It could be from a battery directly too, no matter it is lithium, single 3V button or 2 serial NI-MH battery.

The Power Management Unit (PMU) contains a power clamp, several Low Drop-out Regulators (LDOs), a reference band-gap circuit and temperature sensor. The circuits are optimized for low quiescent current, low drop-out voltage, load regulation, high ripple rejection, and low output noise. The PMU integrates several LDOs for different circuits: RTC LDO, SoC LDO, EXT LDO and INT LDO. They have different operating conditions and features:

- RTC LDO is the main supply only for RTC domain to optimize power consumption at Standby state.

- SoC LDO is the main supply for whole chip digital circuit with programmable voltage from 0.65V to 1.35V to let DVFS operate effectively.
- EXT LDO is main power supply for external device in application, and also can be provide to VDDIO. It has maximum 200mA load current. The output voltage is limited to 3.3V, when VBAT is lower than the value, it will automatically switch to bypass mode to let output voltage follow VBAT.
- INT LDO provide programmable voltage from 0.9V to 2.5V with maximum 600mA load current, for Analog and RF, also SoC LDO input. Normally, make sure VCC voltage is higher than this programmable output voltage setting.

The chip supports to use external DC-DC to further reduce power consumption. After external DC-DC is used, the output voltage on V13 should be at least 0.1V step higher than INT LDO setting voltage to have DC-DC be operating normally.

There are four power domains in the system: RTC domain, AO domain, SoC Switching domain and Radar domain. They mainly used for different scenario to maintain ultra-low power application. We define RS6241 into Active, Idle, Standby and Shutdown power management states to activate different circuit blocks, as shown in Table 7-1.

Table 6-1 Power Management with Block States

Power Mode		CPU	Radar	WCN	EXT /INT LDO	RTC LDO	SoC LDO	DCXO/M SI/DPLL /SPLL	LSI/LSE
Active	All	ON	ON	ON	ON	ON	ON	ON	ON
	CPU	ON	OFF	OFF	ON	ON	ON	OFF/ON/ OFF/ON	ON
	Radar	OFF	ON	OFF	ON	ON	ON	ON	ON
	WCN	ON	OFF	ON	ON	ON	ON	ON/ON/ OFF/OFF	ON
Idle	All	OFF	OFF	OFF	ON	ON	ON	ON	ON
	CPU	OFF	OFF	OFF	ON	ON	ON	OFF/ON/ OFF/ON	ON
	Radar	OFF	OFF	OFF	ON	ON	ON	ON/ON/ ON/OFF	ON
	WCN	OFF	OFF	OFF	ON	ON	ON	ON/ON/ OFF/OFF	ON
Standby		OFF	OFF	OFF	ON	ON	OFF	OFF	ON
Shutdown ¹		OFF	OFF	OFF	ON	ON	OFF	OFF	OFF

1. Chip NRST pin is set to low.

6.5.2 Clock

The clock management system can source the system clocks from a range of internal or external high and low frequency oscillators and distribute them to modules based up the subsystem's individual requirements. The chip has two different kinds of clock source: low frequency clock source(LFCLK) and high frequency clock source(HFCLK).

The chip supports two LFCLK clock sources, the 32.768 KHz crystal oscillator(LSE), LSI(250KHz~2MHz, configurable) The 32.768 KHz crystal oscillator requires an external AT-cut quartz crystal to be connected to the LXL1 and LXL2 pins. The LFCLK is switched off by default when the system is powered up. The LFCLK can be started by selecting the preferred clock source in PRCM register. It is used for each subsystem to achieve lower current consumption for different running mode. In addition, the LFCLK is also used in RTC circuit to achieve accuracy timing.

There are two HFCLK, the 32/40MHz crystal oscillator (XTAL) and 32MHz MSI. The 32MHz MSI is enabled automatically when the system is powered up and can be switched off when all subsystems won't use it anymore in some low power modes. The 32/40MHz XTAL is powered on by CPU after application system is on by software, and controlled by digital signal, so we also name it DCXO in application.

When 2.4GHz wireless subsystem is used, we can only use 32MHz XTAL, while only radar subsystem can use 32MHz or 40MHz XTAL. If higher phase noise performance is required, it is recommended to use 40MHz XTAL for radar.

The HFCLK is used to generate the clock source for SPLL, DPLL WCN SYN, which is used to generate the clock sources for radar, wireless, application system and peripherals. The following figure shows the clock system:

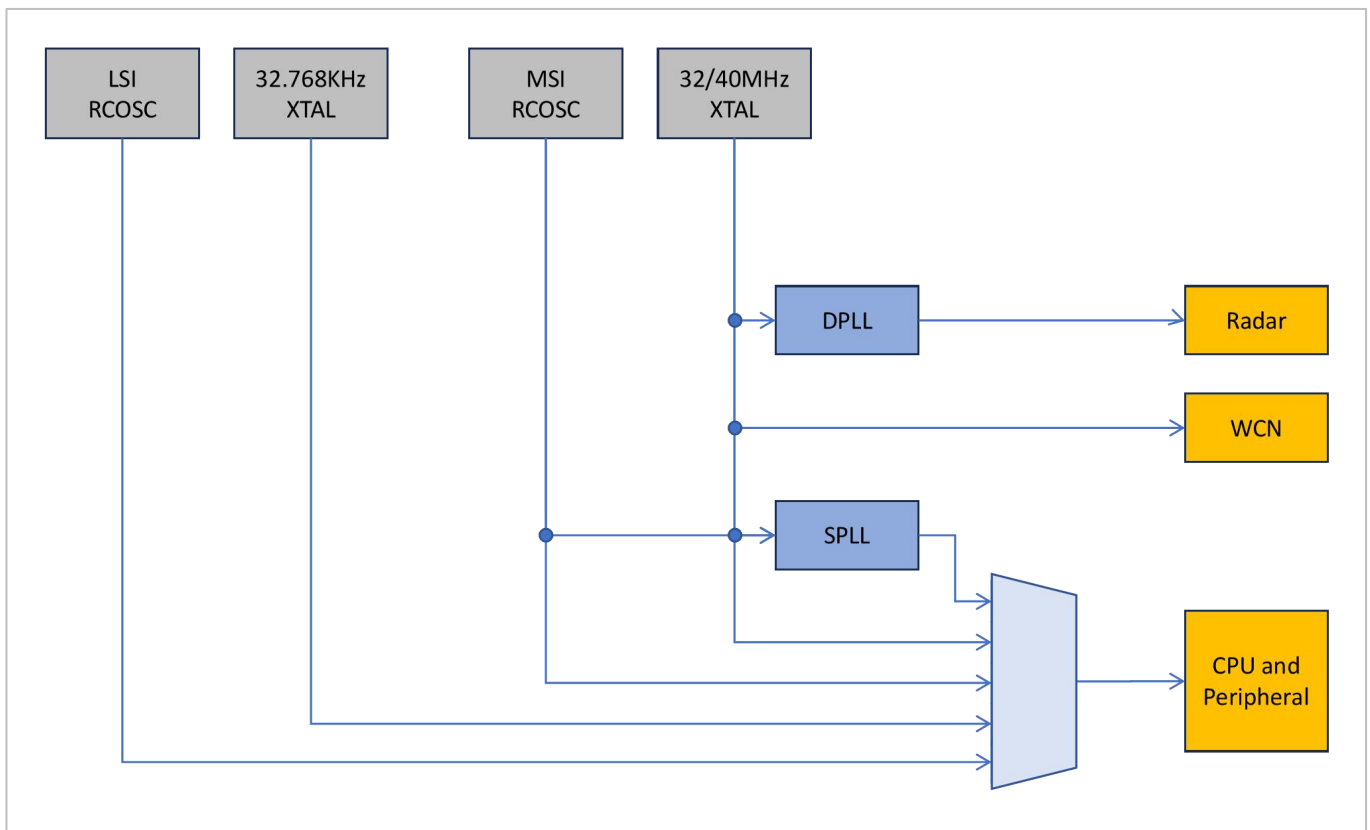


Figure 6-5 Clock System

6.6 Peripherals

6.6.1 GPIO

The RS6241 provides 24 GPIO (General Purpose IO) pins. All ports are mapping out of the device using alternate function multiplexing. The GPIO function can be multiplexed on a multi-function I/O pin by selecting the alternate function in the GPIO Controller Register.

Each GPIO can be configured with the following options:

- Input / Output / Floating(Hi-Z) mode
- Input mode: Pull-up or Pull-down
- Output mode: Active driving@20mA by default, adjustable 20mA per step, up to max 80mA
- Output type: Push-pull or Open-drain
- External Interrupt IO with 5 trigger modes: rising edge, falling edge, double edge, high-level, low-level
- 18 Wakeup IOs can be set to wake system by external interrupt at Standby state
- All IOs can be set to wake system by external interrupt in Idle state

8 IOs are multiplexed with 8 channels ADC. A dedicated internal control signal is used to select between the digital and analog functions.

6.6.2 UART

The RS6241 have three embedded UART controllers: UART2 is used for debug, UART0 and UART1 with auto-flow control are used for communication with external devices. UART0 is used for system boot configuration and firmware upgrade by default. The UART has industry-standard 16550 mode of operation.

Features:

- Compatible with industry-standard 16550
- Support DMA controller interface
- 64-Bytes Transmit and Receive data FIFO
- Support Programmable fractional baud rate
- Support Hardware auto-flow control
- Support baud rate detection
- Support RS-232
- Support clock detection to wake up system from Standby state

6.6.3 QSPI

The RS6241 embeds a Quad Serial Peripheral Interface (QSPI). The QSPI protocol supports half-duplex, full-duplex, synchronous and serial communication with peripheral device. The interface can be configured as master and slave. There are four possible combinations for the serial clock phase and polarity.

Features:

- Clock frequency up to 96MHz
- Support Motorola SPI mode

- Programmable Master or Slave
- Support Standard SPI, Dual SPI and Quad SPI mode
- Programmable Data width size 4 to 32 bits
- 32 words FIFO depth for both Transmit and Receive data
- Programmable sample delay time in master mode
- Programmable Serial Clock Polarity (SCPOL) and Serial Clock Phase (SCPH)
- Support Interrupt and DMA
- Programmable Instruction, Address Length and Wait Cycle in Master mode
- Support clock detection to wake up system from Standby state

6.6.4 I²C

The RS6241 embeds two two-wire serial interfaces (I²C). Each I²C can be configured as master or slave mode. The I²C can operate in standard mode (with data rate 0 to 100 Kb/s), fast mode (with data rates less than or equal to 400 Kb/s), fast mode plus (with data rates less than or equal to 1000 Kb/s).

Features:

- Programmable Master or Slave
- Support three speeds: Standard mode (0 to 100 Kb/s), Fast mode (≤ 400 Kb/s), Fast mode plus (≤ 1000 Kb/s)
- Support 7- or 10bit addressing
- Support Multi-master system
- 8 bytes FIFO depth for Transmit and Receive data
- Programmable SDA hold time
- Support DMA
- Support Address detection to wake up system from Standby mode

6.6.5 PWM

The RS6241 has 4 PWM generators, every PWM generator have a separate counter to output 2 channels. Each PWM generator is a complementary pair output. Only PWM 0/2/4/6 channels are provided with capture mode. The capture detects the rising edge and the falling edge of the signal and calculate capture counter to compare match value.

Features:

- 4 PWM generators, 8 output channels and 4 input capture channels
- Support input capture and interrupt

- Programmable dead-band
- Support pulse, period and complementary pair outputs
- PWM generation with full modulation capability (0-100%)

6.6.6 Audio PWM

The PWM Audio is a high-efficiency mono Class-D Audio Amplifier. The digital audio interface accepts 32 kHz, 44.1 kHz and 48 kHz sample rate with 16-/32-bit data supported for all data formats. The PWM Audio pins is multiplexed with PWM.

Features:

- Support mono mode
- Configurable sample rates: 32 kHz, 44.1 kHz, 48 kHz
- 64 FIFO depth for mono mode
- Support Volume control
- Support interrupt

6.6.7 GPADC

The GPADC contains a 10-channel analog switch, a single end input asynchronous 12-bit SAR ADC. The channels 0 to 7 are used to detect the voltage of the external input, The channel 8 is used to detecting the voltage of VINT, and the channel 9 is dedicated to detecting the voltage of the VCC.

Features:

- 12-bit resolution and 10-bit effective SAR type A/D converter
- 10-channel multiplexer, 8 normal channels and 2 VCC voltage detection channel
- Up to 32 FIFO depth for data
- Support DMA
- Support calibration and compensation
- Internal Vref 1.2V, Support five class of input voltage attenuation: 2/3, 1/2, 1/3, 1/5, 1/9. The max voltage is 3.6V
- Maximum sampling frequency: 2 MHz
- Programmable oversampling: 16, 32, 64, 128
- Support one channel 4096 oversampling
- Support compare and data interrupt
- Support four operation modes: single mode, single-cycle mode, continuous mode, burst mode

- Support hardware triggers (internal timer periodic event or WIO input event) to start conversion in STANBY mode. It generates an interrupt to wake system when data is over the range of the comparison window.

6.6.8 CAN-FD

The CAN controller supports CAN2.0B, CAN FD, TTCAN. contains configurable receive buffer, two transmit buffers, four acceptance filters.

Features:

- CAN2.0B: up to 8 bytes payload, data rates up to 1Mbit/s
- CAN FD: up to 64 bytes payload, data rate limited by the transceiver and the clock frequency of the CAN-CTRL core
- TTCAN: support Time-Triggered CAN which can be combined with CAN2.0B or CAN FD
- Configurable receive buffer size: three buffer slots, FIFO-like behavior
- Two transmit buffers: one primary transmit buffer, three secondary transmit buffer
- Four acceptance filters: acceptance code can be checked by bit
- Extended feature: single shot transmission mode; listen only mode

6.6.9 LIN

The LIN controller support following features:

- Supports transmission rates up to 1 MHz.
- Supports host mode for sending and receiving messages with different frame IDs.
- Supports slave mode for responding to and processing messages with different IDs.
- Supports arbitration adaptation in multi-slave environments.
- Adjustable separation field lengths.
- Supports both traditional checksum and enhanced checksum verification modes.
- In host mode, different frame slot widths can be set

7 Package Description

7.1 Package Outline

The RS6241 chip-set is housed in the FCCSP package with bump balls of 350µm diameter and standoff of 250 µm. The package size is (6.7 x 6.7 x 0.87) mm³ with a general ball pitch of 650 µm.

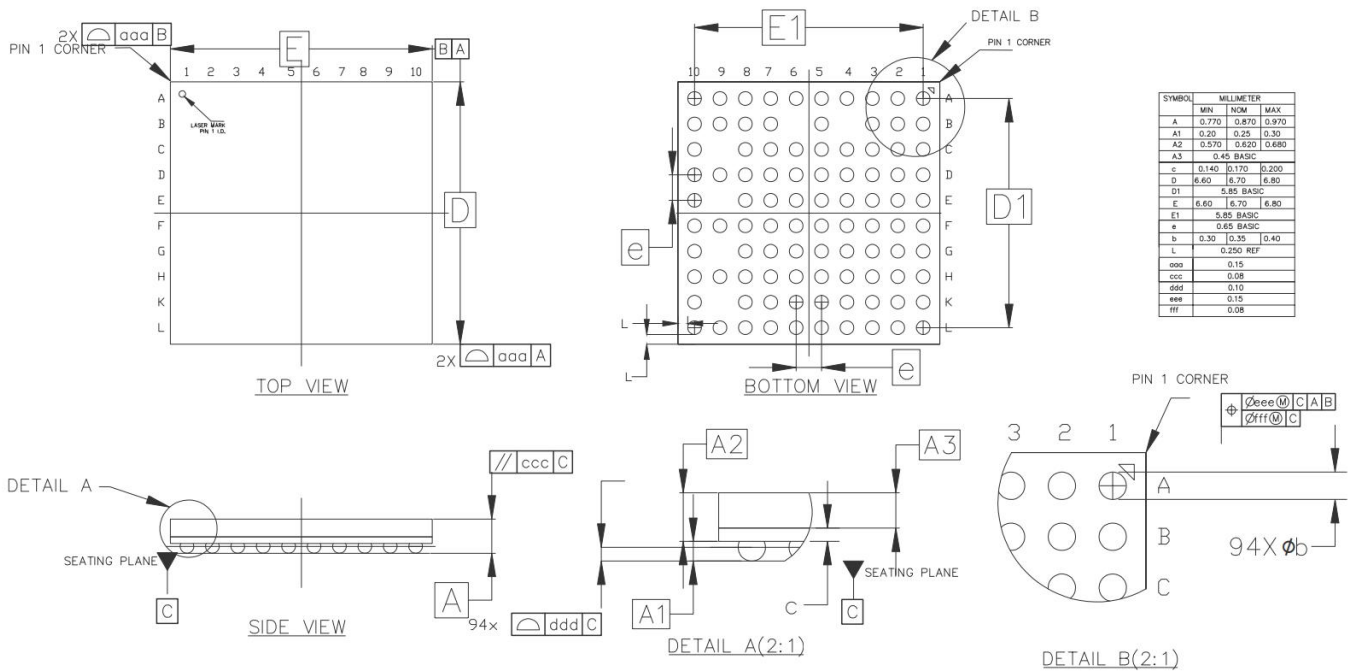


Figure 7-1 Package Outline Diagram

7.2 PCB Land Pattern

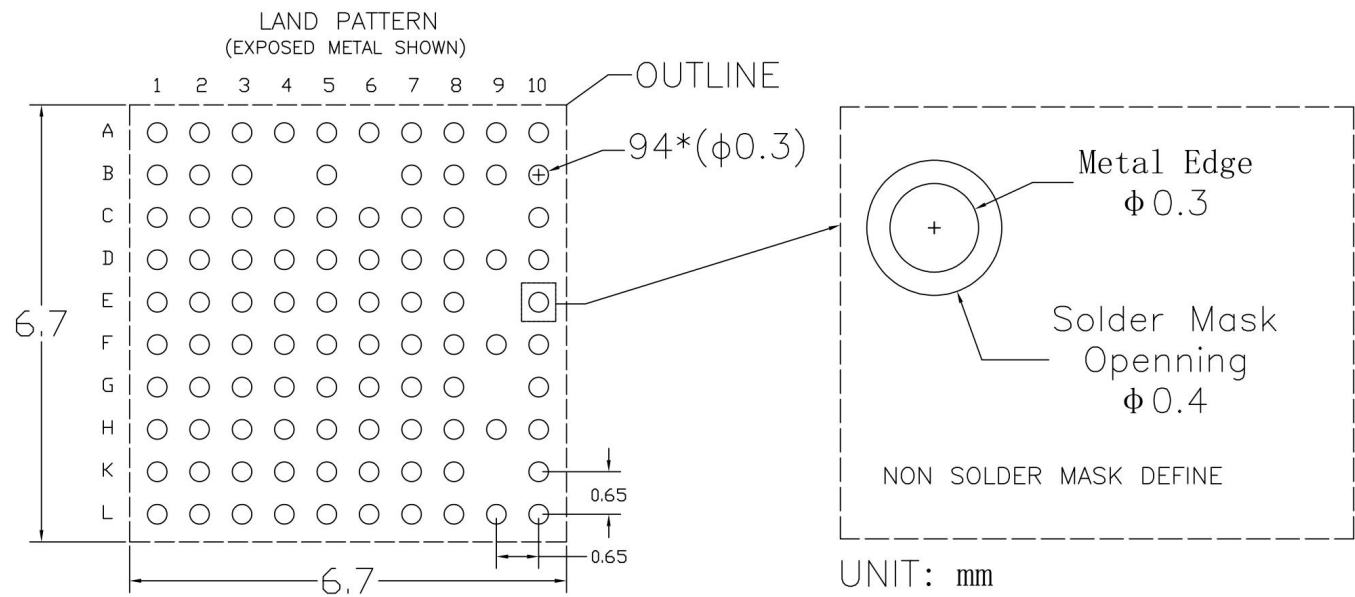


Figure 7-2 PCB Land Pattern Diagram

7.3 Package Thermal Property

Table 7-1 Thermal Property

Thermal Parameters		°C/W
$R\theta_{JC}$	Junction-to-case	TBD
$R\theta_{JB}$	Junction-to-board	TBD
$R\theta_{JA}$	Junction-to-free air	TBD
Ψ_{iJA}	Junction-to-package top	TBD
Ψ_{iJB}	Junction-to-board	TBD

8 Application Circuit

8.1 Schematic Notes

- **POWER**

- **OPTION A: “DCDC”** section will help to reduce power consumption compared to using RS6241 Inner-LDO, if application don't care this, can delete it to save BOM cost.
- VCC and VDDIO are powered by different external power source, however, when VCC is below 3.6V, VCC and VDDIO could share the same power source.

- **BLE**

- **OPTION B: “32.768KHz”** section is used for BLE system, if application doesn't contain BLE, can delete it.
- **OPTION C: “BLE ANTENNA”** section is used for BLE system, if application doesn't contain BLE, can delete it.

- **EXTERNAL CONNECT**

HOST MCU could control the radar through any interface in UART0/I2C0/SPI0, and UART0(PB0/PB1) is used by default, PA0/PA1 is used as radar interrupt signal. NRST is recommended to control by host mcu when VCC hasn't power switch.

- **OPTION D:** CONNECT By I2C0(PA4/PA5), radar interrupt signal(PA0/PA1).
- **OPTION E:** CONNECT By SPI0(PA0/PA1/PA2/PA3), radar interrupt signal(PA6/PA7).

It is strongly recommended to reserve PA0, PB0, and PB1 as test points, as these three signals will be used for noise testing, it will be convenient for radar performance analysis.

- **PSRAM**

- **OPTION F: “PSRAM”** is used as external memory, if application doesn't require functions such as posture detection or high-performance human presence sensing that require computing large amounts of data, can delete it. It is recommended to reserve it.

8.2 Reference Schematic

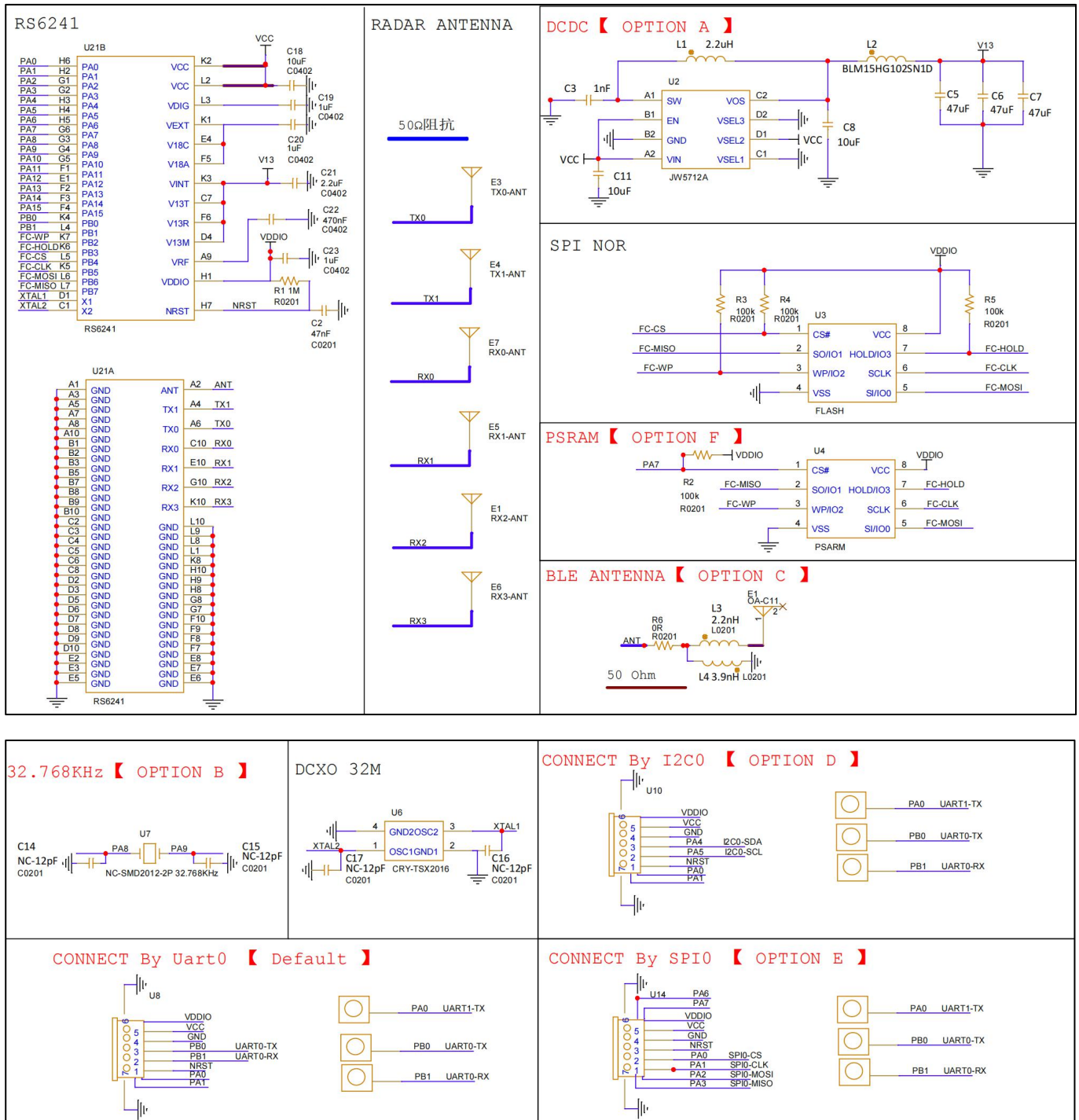


Figure 8-1 Reference Schematic

9 Silkscreen

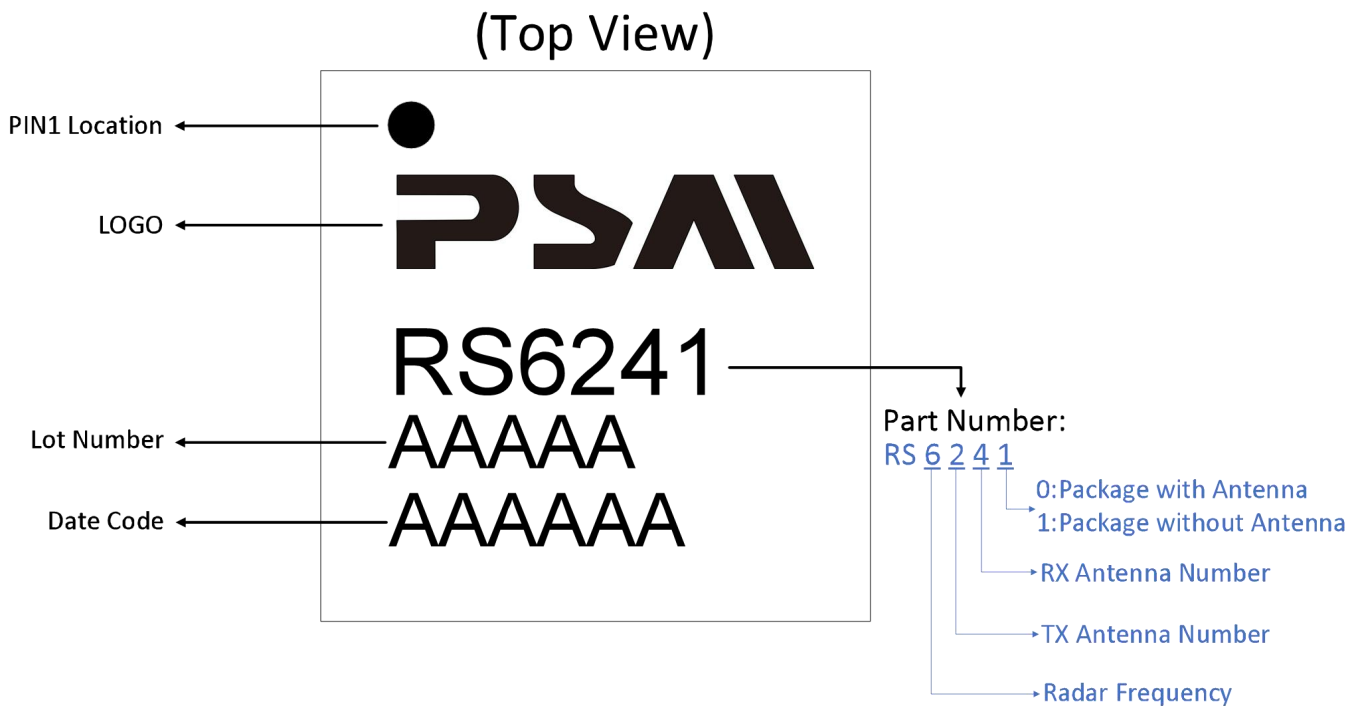


Figure 9-1 Silkscreen

10 Shipping package

Unit: mm

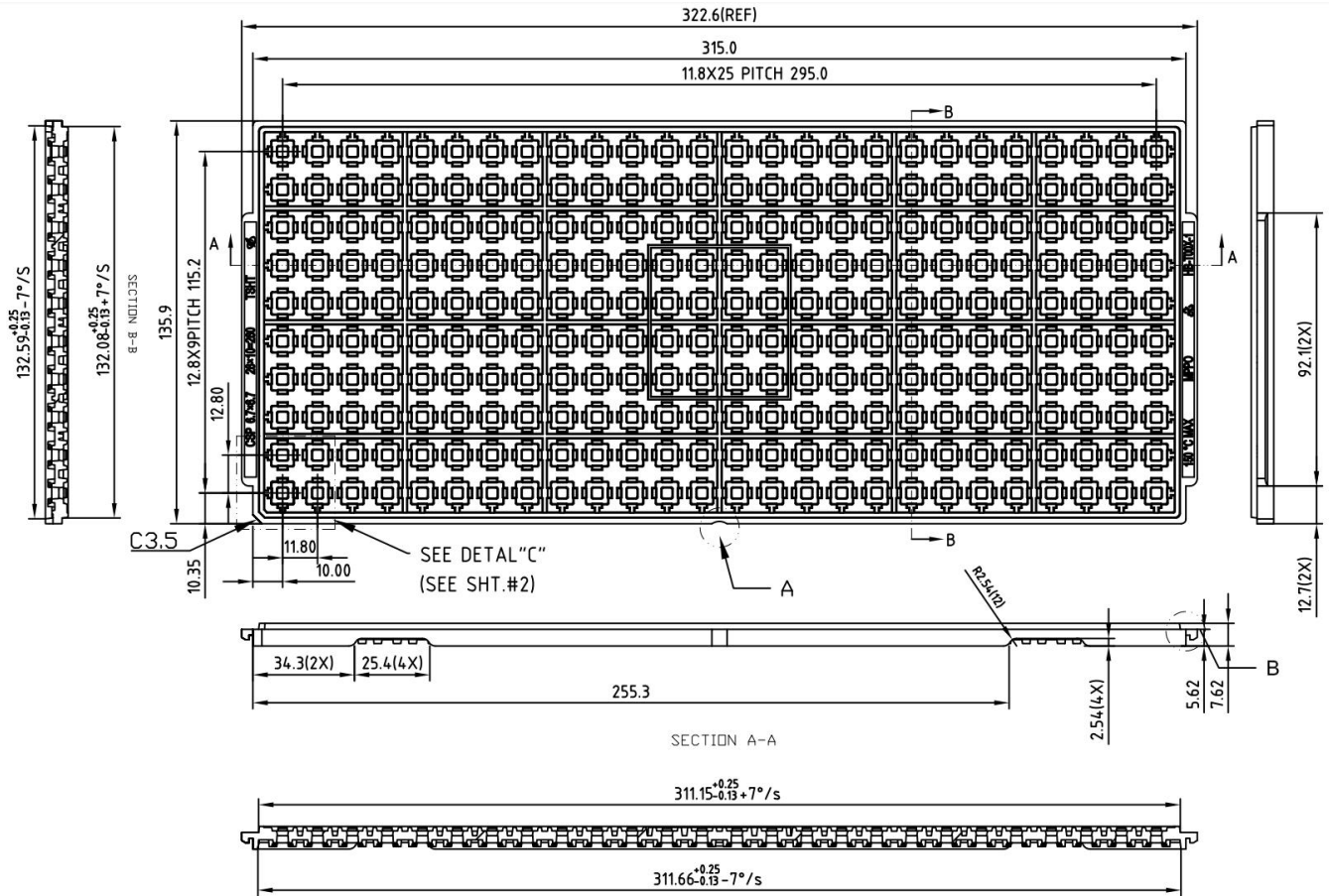


Figure 10-1 Shipping package

11 Reflow Soldering

Suggested peak reflow temperature is 240 °C ,not exceeding 15 seconds.The recommended reflow soldering heat curve(lead-free reflow soldering)and related parameters are as follows:

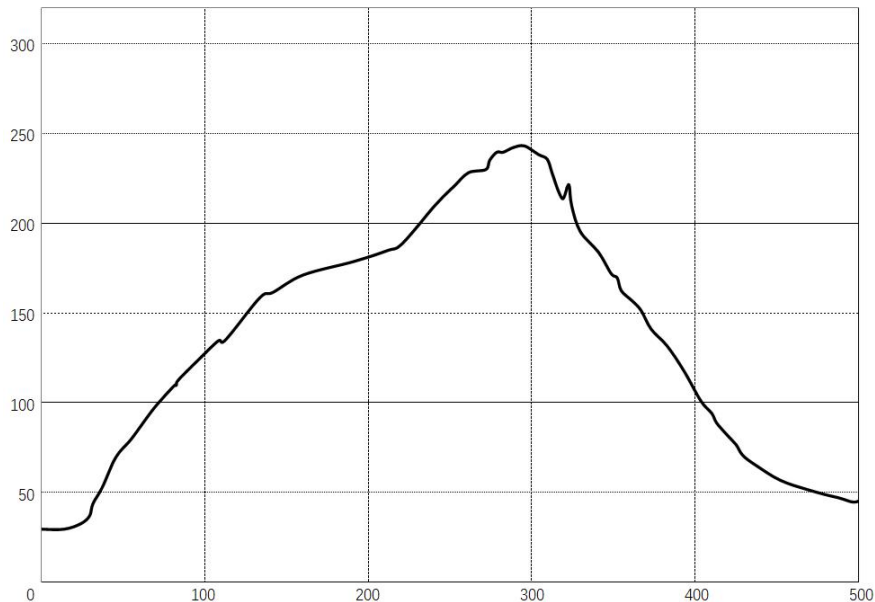


Figure 11-1 Recommended reflow soldering heat curve

Table 11-1 Recommended thermal distribution parameters

Parameter	Recommended value
Max. slope for heating up	2.0~4.0°C/sec
Max. slope for cooling down	1.0~5.0°C/sec
Soak time (150°C~ 200°C)	60~120 sec
Reflow time (over 220°C)	30~90 sec
Max. temperature	240°C

Disclaimer and Copyright Statement

Our company strives to provide accurate information in this document, but we cannot guarantee that the complete content is correct. Damages (including but not limited to indirect, incidental, special losses) may occur due to the use of this document, or failure to comply with the instructions for use of this document. Our company is not responsible for any losses caused by improper use (including but not limited to over voltage, over clocking, and over temperature use).

The content in this text, including but not limited to the URL address provided for reference, may differ from the actual content due to product version upgrades or other reasons, and is subject to change without prior notice. This document is provided "as is" without any warranty of any kind, including any warranty of merchant ability, fitness for a particular purpose or purpose, and any proposals, specifications or samples mentioned in any warranty.

This document does not assume any liability, including liability for infringement of any patent rights arising from the use of the information in this document to add value to the industry. This document does not convey any license, express or implied, to any intellectual property rights by estoppel or otherwise.

Copyright©2026 Possumic Technology Co., Ltd. all rights reserved.